

# Computer Science & IT

## Discrete Mathematics



### Graph Theory

Lecture No. 08



By- Vishal Sir



# Recap of Previous Lecture

- ✓ **Topic** Matching
- ✓ **Topic** Line covering
- ✓ **Topic** Line independent set





# Topics to be Covered



Topic

Vertex covering



Topic

Vertex independent set



Topic

Connectivity

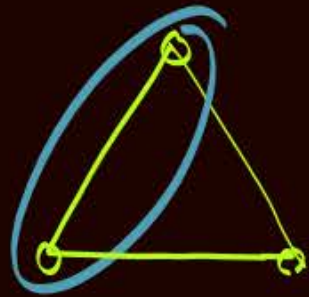


H.W. Q:- Find the matching number of Complete graph  $K_n$



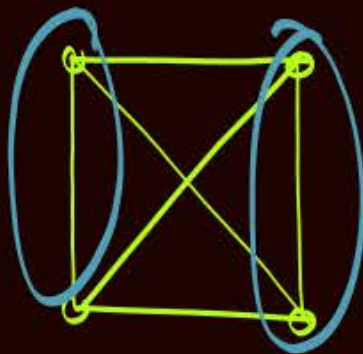
$K_2$

Matching No.  
of  $K_2 = 1$



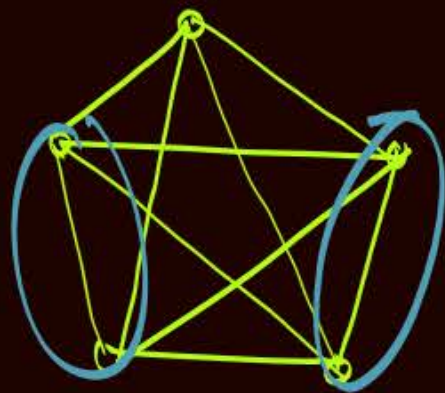
$K_3$

Matching No.  
of  $K_3 = 1$



$K_4$

Matching No.  
of  $K_4 = 2$



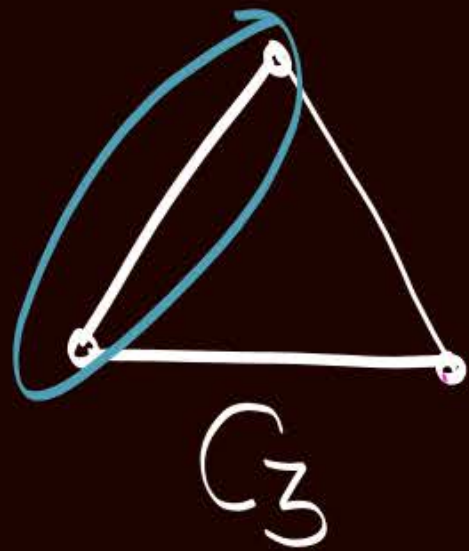
$K_5$

Matching No.  
of  $K_5 = 2$

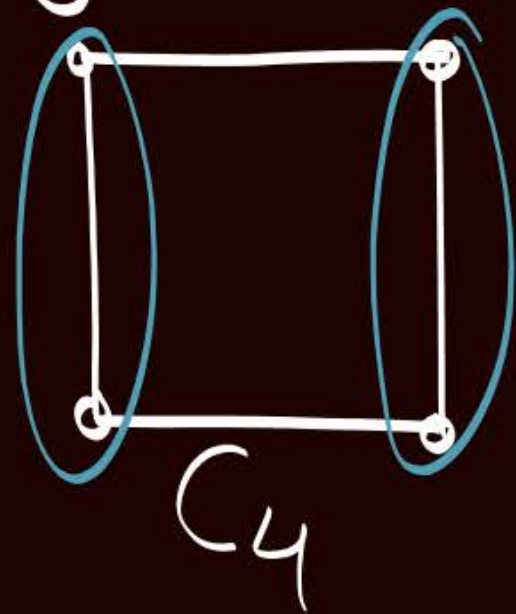
$$\text{Matching no. of } K_n = \left\lfloor \frac{n}{2} \right\rfloor$$



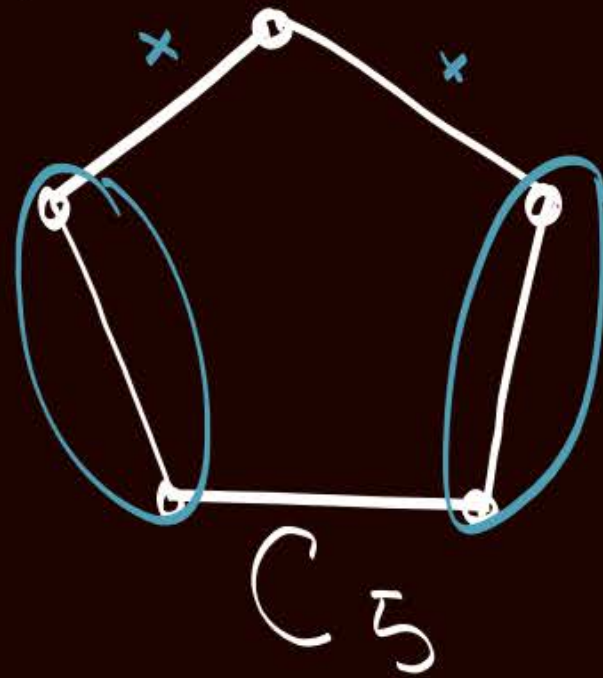
H.W.  
Q: Find matching number of Cycle graph  $C_n$



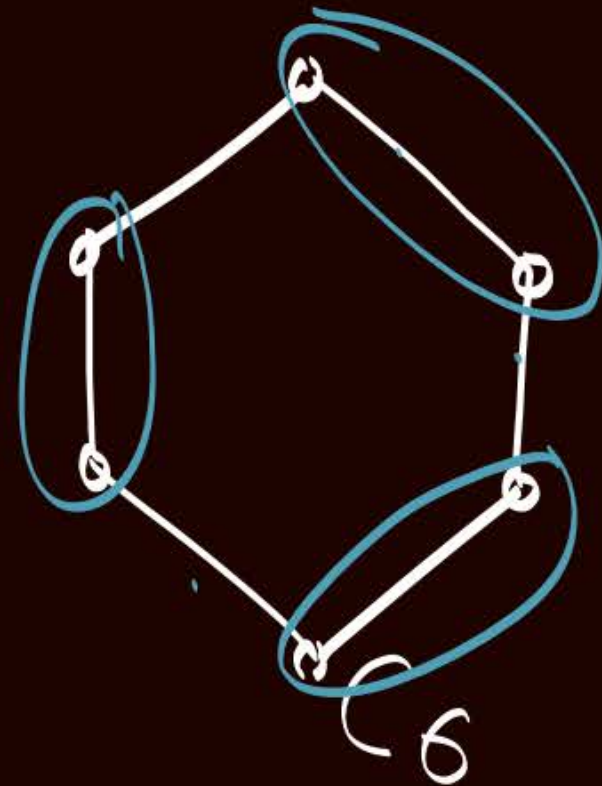
Matching No. = 1



Matching No. = 2



Matching No. = 2

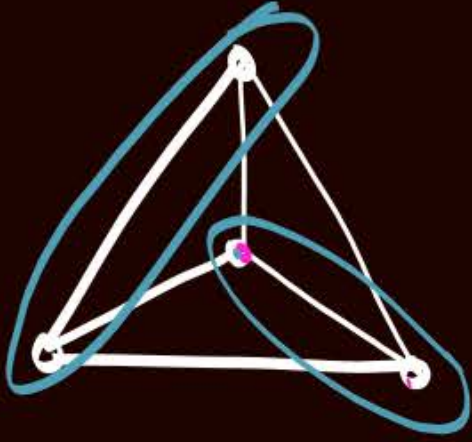


Matching No. = 3

Matching no. of  $C_n = \lfloor \frac{n}{2} \rfloor$

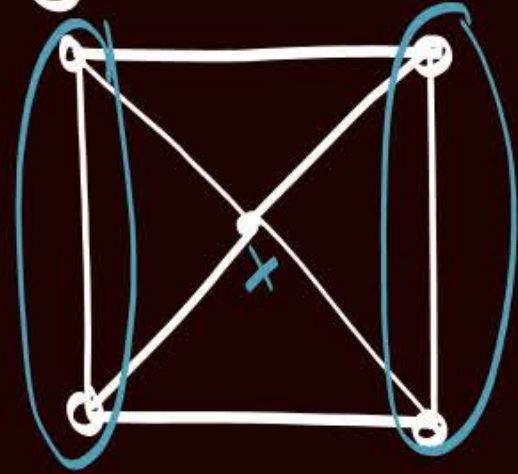
H.W.

Q: Find matching number of wheel graph  $W_n$



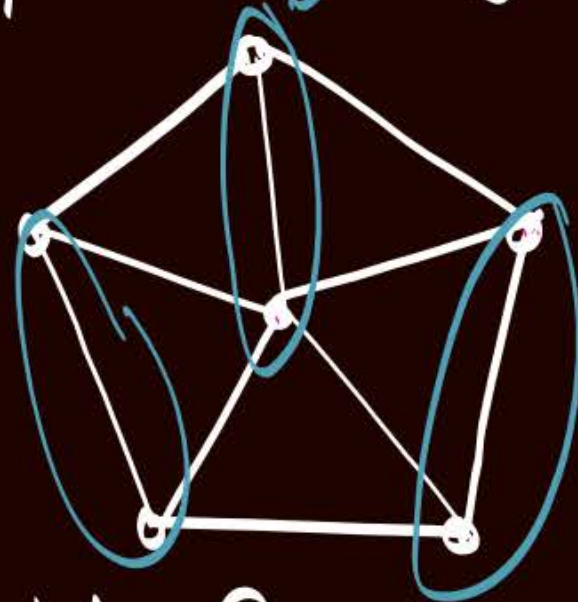
$$W_4 = C_3 + 1$$

Matching No. = 2



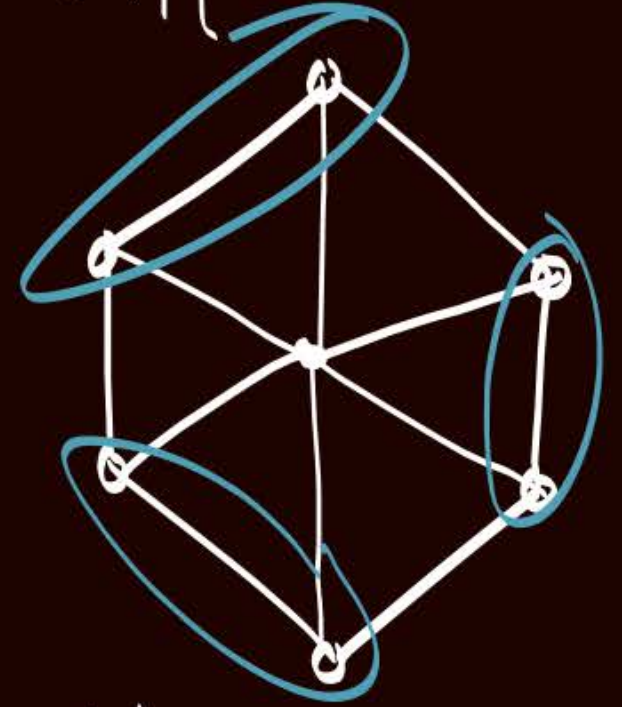
$$W_5 = C_4 + 1$$

Matching No. = 2



$$W_6 = C_5 + 1$$

Matching No. = 3



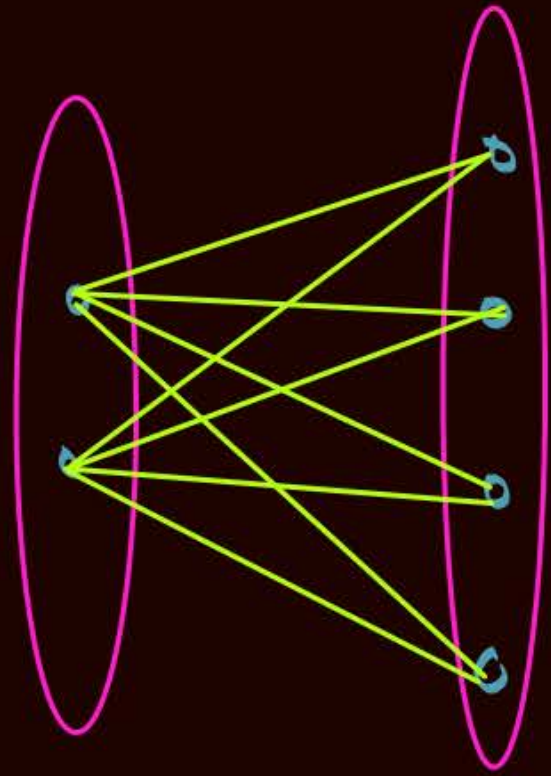
$$W_7 = C_6 + 1$$

Matching No. = 3

$$\text{Matching no. of } W_n = \left\lfloor \frac{n}{2} \right\rfloor$$

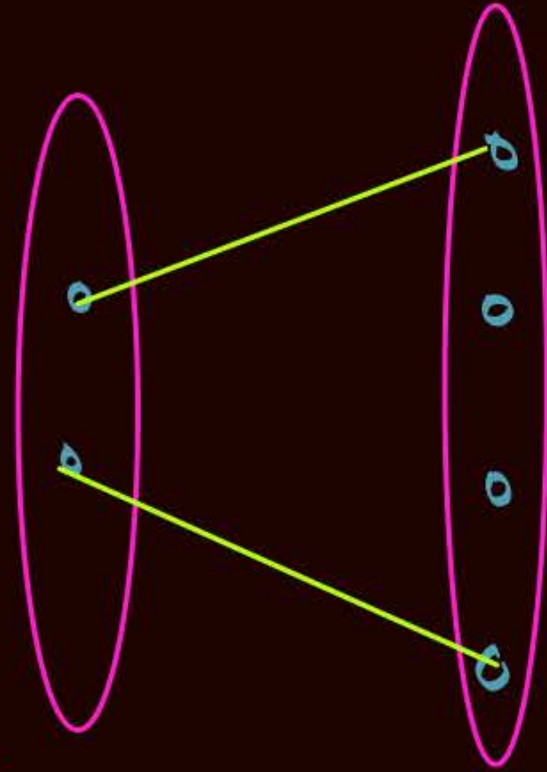


H.W Q:- Find the matching number of Complete bipartite graph  $K_{m,n}$



$K_{2,4}$

Maximum  
matching



Matching no. = 2

Matching no. of  
 $K_{m,n} = \text{Min}(m,n)$

Q:- Find the matching number of star graph with 'n' vertices.

$$L \equiv K_{1,n-1} \text{ or } K_{n-1,1}$$

$$\therefore \text{Matching no.} = \text{Min}(1, n-1) \\ = 1$$



# Covering

```
graph TD; Covering[Covering] --> LineCovering[Line Covering / Edge Covering]; Covering --> VertexCovering[Vertex Covering]; LineCovering --> LineCoveringText[We will try to Cover all the vertices of the graph using Subset of edges]; VertexCovering --> VertexCoveringText[We will try to Cover all the edges of the graph using Subset of vertices];
```

Line Covering / Edge Covering

We will try to Cover all the vertices of the graph using Subset of edges

Vertex Covering

We will try to Cover all the edges of the graph using Subset of vertices



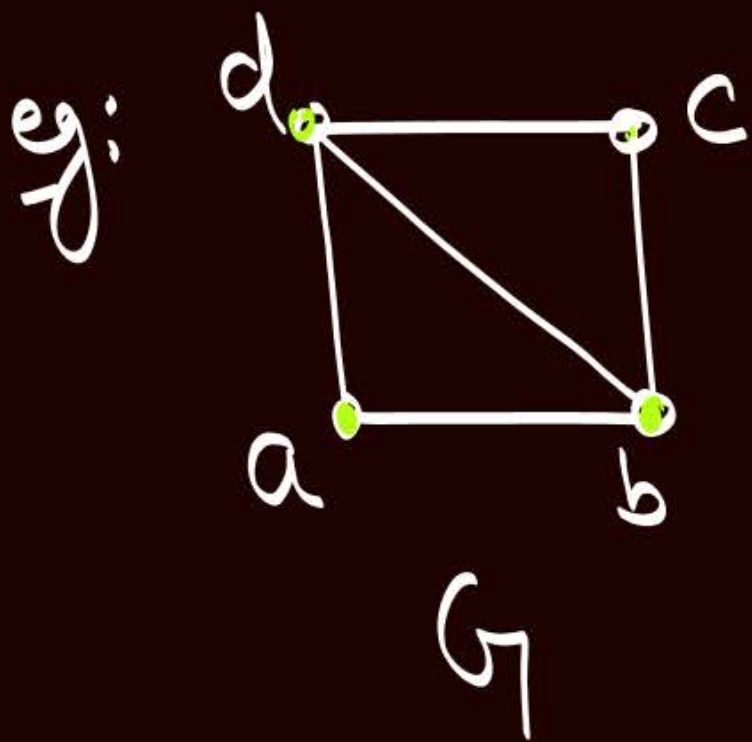
## Topic : Vertex Covering

Let  $G=(V,E)$  be a graph.

A subset  $K$  of set of vertices  $V$  is called vertex covering of graph  $G$  if every edge of the graph is incident with at least one vertex of set  $K$

{ i.e, Every edge of graph  $G$  must have at least one of its end vertices in subset  $K$  }





$$K_1 = \{a, b, c, d\} \quad \times$$

$$\checkmark K_2 = \{a, c, b\}$$

$$\checkmark K_3 = \{a, c, d\}$$

$$\checkmark K_4 = \{b, d\}$$

$$K_5 = \{a, b, d\}$$

All subsets of  
Vertices from  
 $K_1$  to  $K_5$   
are vertex  
Covering of graph  $G$



## Topic : Minimal vertex covering ✓

- A vertex covering of graph  $G$  is called minimal vertex covering if no vertex can be deleted from the set without destroying its ability to cover all the edges.

In the above example,

$K_2$ ,  $K_3$  &  $K_4$  are minimal vertex covering





## Topic : Minimum vertex covering

✓ Smallest minimal vertex covering

→ Vertex coverings of graph  $G$  with minimum number of vertices are called minimum vertex coverings of graph  $G$ .

In the above example,

$K_4$  is the minimum vertex covering

\* Every minimum vertex covering is also a minimal vertex covering but Every minimal vertex covering need not be a minimum vertex covering (eg:  $K_2$  &  $K_3$ )





## Topic : Vertex covering number ( $\alpha_2$ )

Vertex Covering number  
of graph  $G$  ( $\alpha_2$ ) = No. of Vertices in any one of  
the minimum Vertex Covering of  $G$

In the above example,

$$\alpha_2 = 2$$

(wrt.  $K_4$ )



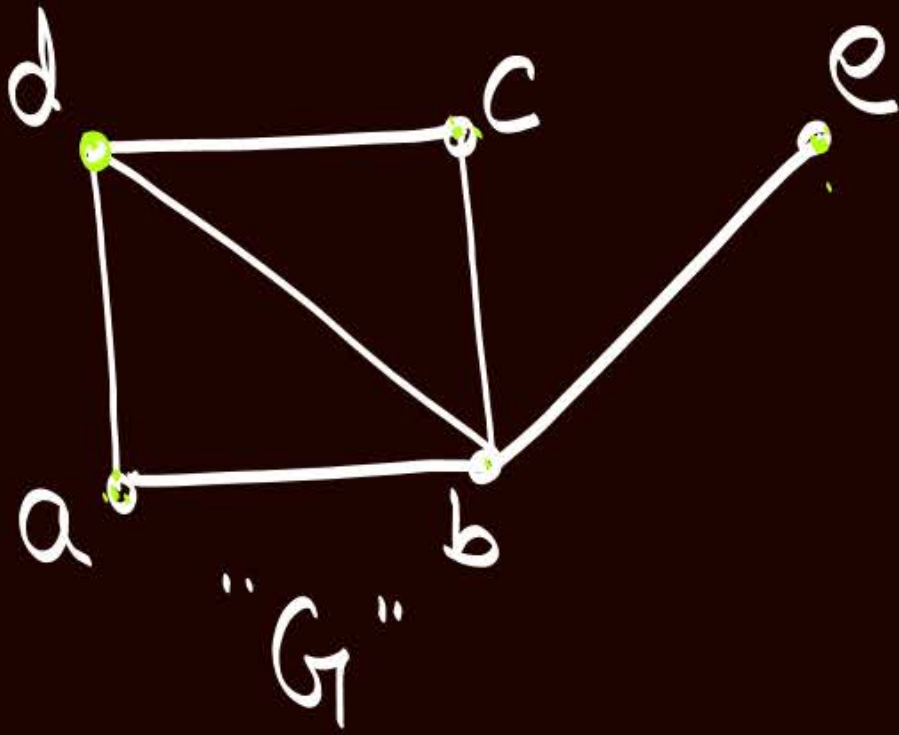
## Topic : Vertex independent set

→ Let  $G = (V, E)$  be a graph.

A subset 'S' of set of vertices  $V$  is called a vertex independent set if no two vertices of subset  $S$  are adjacent to each other.



eg:



$$S_1 = \{e\}$$

$$S_2 = \{b\}$$

$$S_3 = \{a, e\}$$

$$S_4 = \{a, c, e\}$$

$$S_5 = \{d, e\}$$

A subset of vertices with a single vertex is always a vertex independent set.

All subset of vertices from  $S_1$  to  $S_5$  are vertex independent set w.r.t. graph  $G$ .



## Topic : Maximal vertex independent set

→ A vertex independent set of graph  $G$  is called a maximal vertex independence set if no new vertex of the graph can be added to the set without destroying its property of being independent.

In the above example.

$S_2$ ,  $S_4$  &  $S_5$  are maximal vertex independent set.





## Topic : Maximum vertex independent set



Largest maximal vertex independent set

- Vertex independent sets with maximum number of vertices are called maximum vertex independent set of graph  $G$

In the above example,

' $S_4$ ' is maximum vertex independent set.

\* Every maximum vertex independent set is also a maximal vertex independent set, but

Every maximal vertex independent set need

not be a maximum vertex independent set. eg.  $\{S_2, S_5\}$





## Topic : Vertex independence number ( $\beta_2$ )

Vertex independence Number ( $\beta_2$ ) of graph  $G$  = No. of vertices in any one of the maximum vertex independent set of graph  $G$ .

In the above example,

$$\beta_2 = 3$$

"w/o  $S_4$ "



## Topic : NOTE



For any graph  $G = (V, E)$

$$\alpha_2 + \beta_2 = |V(G)|$$

(Vertex Covering no. of graph  $G$  + Vertex independence no. of graph  $G$ ) = Total no. of Vertices in graph  $G$



Find  $\alpha_1$ ,  $\beta_1$ ,  $\alpha_2$  &  $\beta_2$  for the following graphs

- (i) Complete graph  $K_n$
- (ii) Cycle graph  $C_n$
- (iii) Wheel graph  $W_n$
- (iv) Complete bipartite graph  $K_{m,n}$
- (v) Star graph with 'n' vertices

$\alpha_1$  = line covering no.

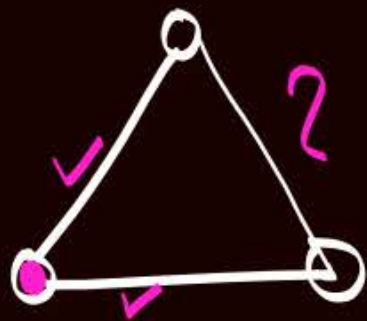
$\beta_1$  = line independence no.

$\alpha_2$  = Vertex covering no.

$\beta_2$  = vertex independence no.

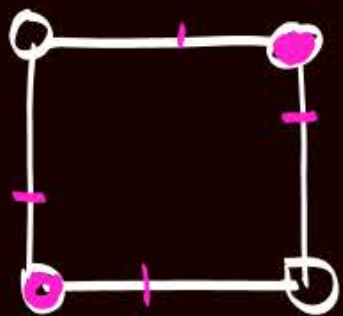
|                                          | Line Covering No.<br>$\alpha_1$ | Line independence No.<br>$\beta_1$ | Vertex Covering No.<br>$\alpha_2$                               | Vertex independence No.<br>$\beta_2$ |
|------------------------------------------|---------------------------------|------------------------------------|-----------------------------------------------------------------|--------------------------------------|
| ① Complete graph ( $K_n$ )               | $\lceil \frac{n}{2} \rceil$     | $\lfloor \frac{n}{2} \rfloor$      | $(n-1)$                                                         | 1                                    |
| ② Cycle graph ( $C_n$ )                  | $\lceil \frac{n}{2} \rceil$     | $\lfloor \frac{n}{2} \rfloor$      | $\lceil \frac{n}{2} \rceil$                                     | $\lfloor \frac{n}{2} \rfloor$        |
| ③ Wheel graph ( $W_n$ )                  | $\lceil \frac{n}{2} \rceil$     | $\lfloor \frac{n}{2} \rfloor$      | $1 + \lceil \frac{n-1}{2} \rceil = \lceil \frac{n+1}{2} \rceil$ | $\lfloor \frac{n-1}{2} \rfloor$      |
| ④ Complete Bipartite graph ( $K_{m,n}$ ) | $\text{Max}(m, n)$              | $\text{Min}(m, n)$                 | $\text{Min}(m, n)$                                              | $\text{Max}(m, n)$                   |
| ⑤ Star graph with $n$ -vertices          | $(n-1)$                         | 1                                  | 1                                                               | $(n-1)$                              |





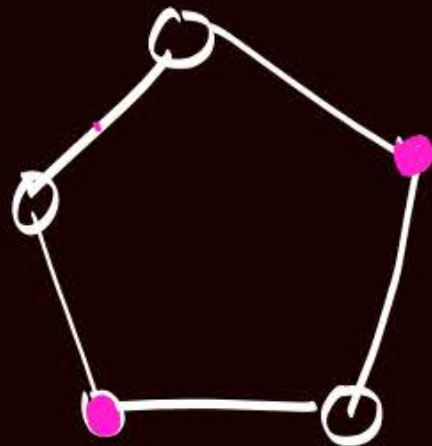
$$\alpha_2 = 2$$

$$\beta_2 = 1$$



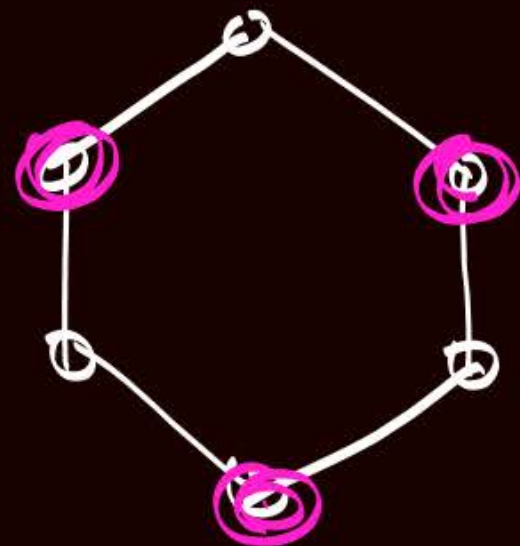
$$\alpha_2 = 2$$

$$\beta_2 = 2$$



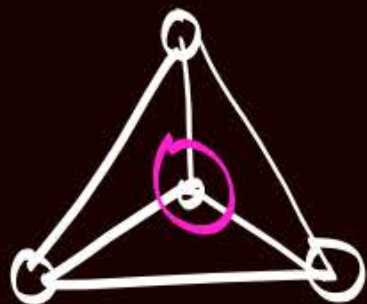
$$\alpha_2 = 3$$

$$\beta_2 = 2$$

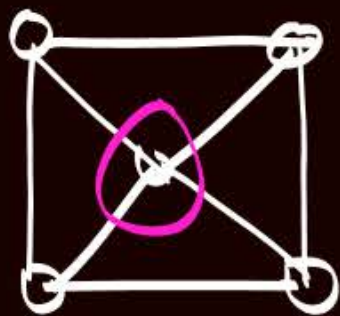


$$\alpha_2 = 3$$

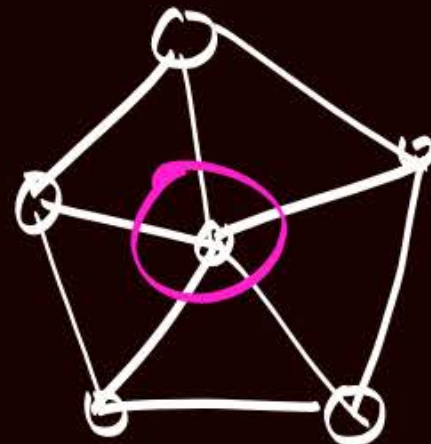
$$\beta_2 = 3$$



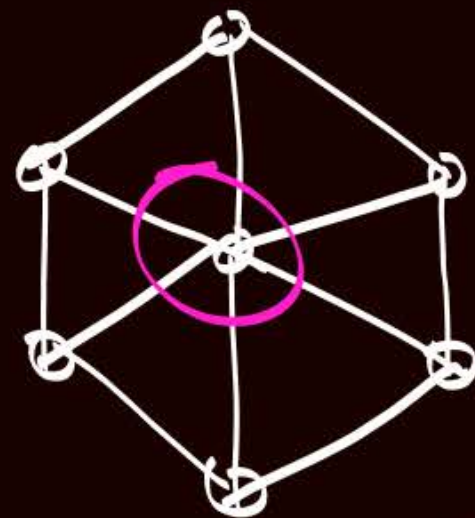
$$W_4 = C_3 + 1$$



$$W_5 = C_4 + 1$$



$$W_6 = C_5 + 1$$



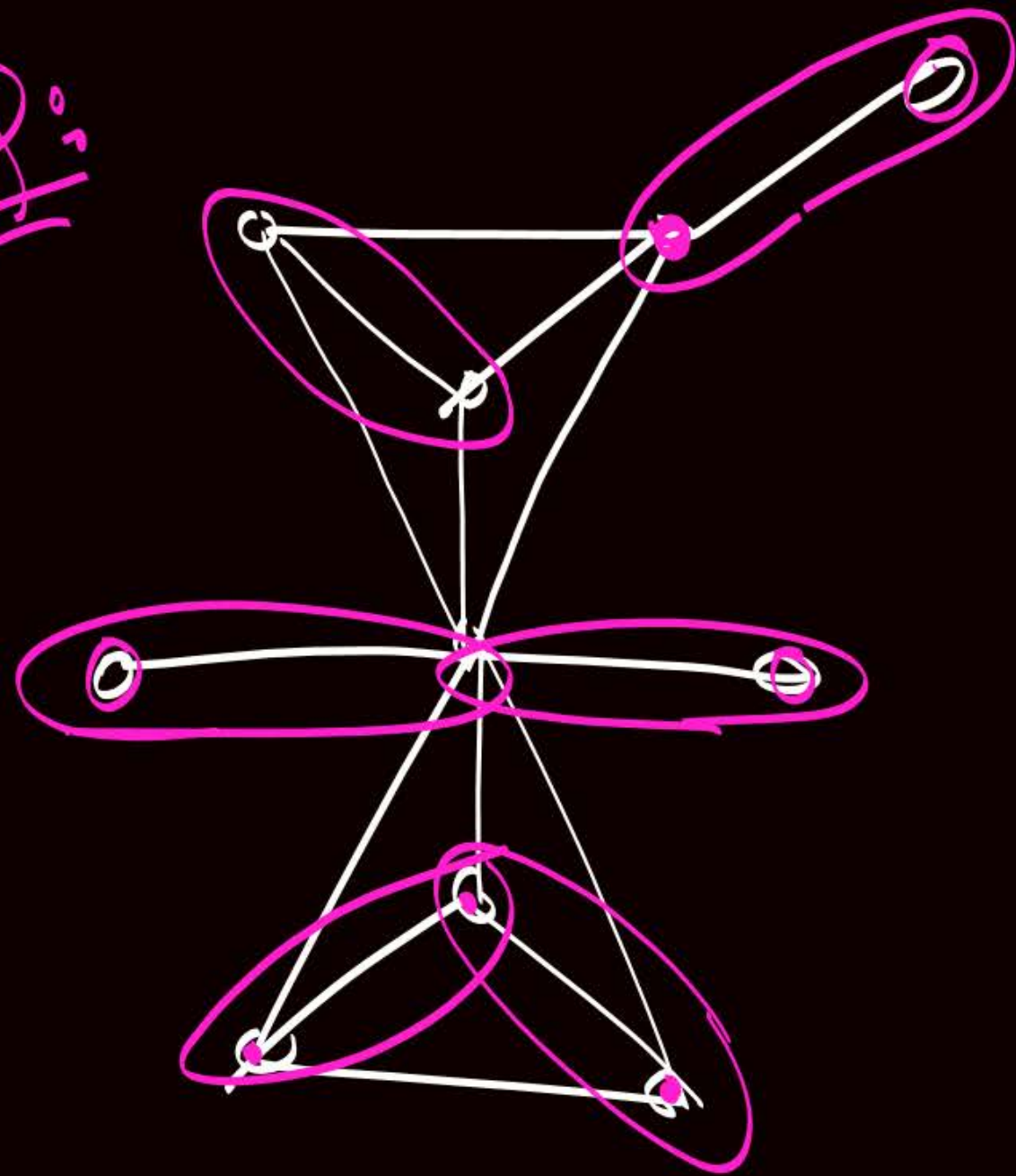
$$W_7 = C_6 + 1$$

$$\beta_2(W_n) = \beta_2(C_{n-1}) = \left\lfloor \frac{n-1}{2} \right\rfloor$$

$$\alpha_2(W_n) = \underset{\substack{\uparrow \\ \text{for all} \\ \text{spokes} \\ \text{of wheel}}}{1 \text{ hub}} + \alpha_2(C_{n-1}) = 1 + \left\lfloor \frac{n-1}{2} \right\rfloor = \left\lfloor \frac{n+1}{2} \right\rfloor$$



Q.11

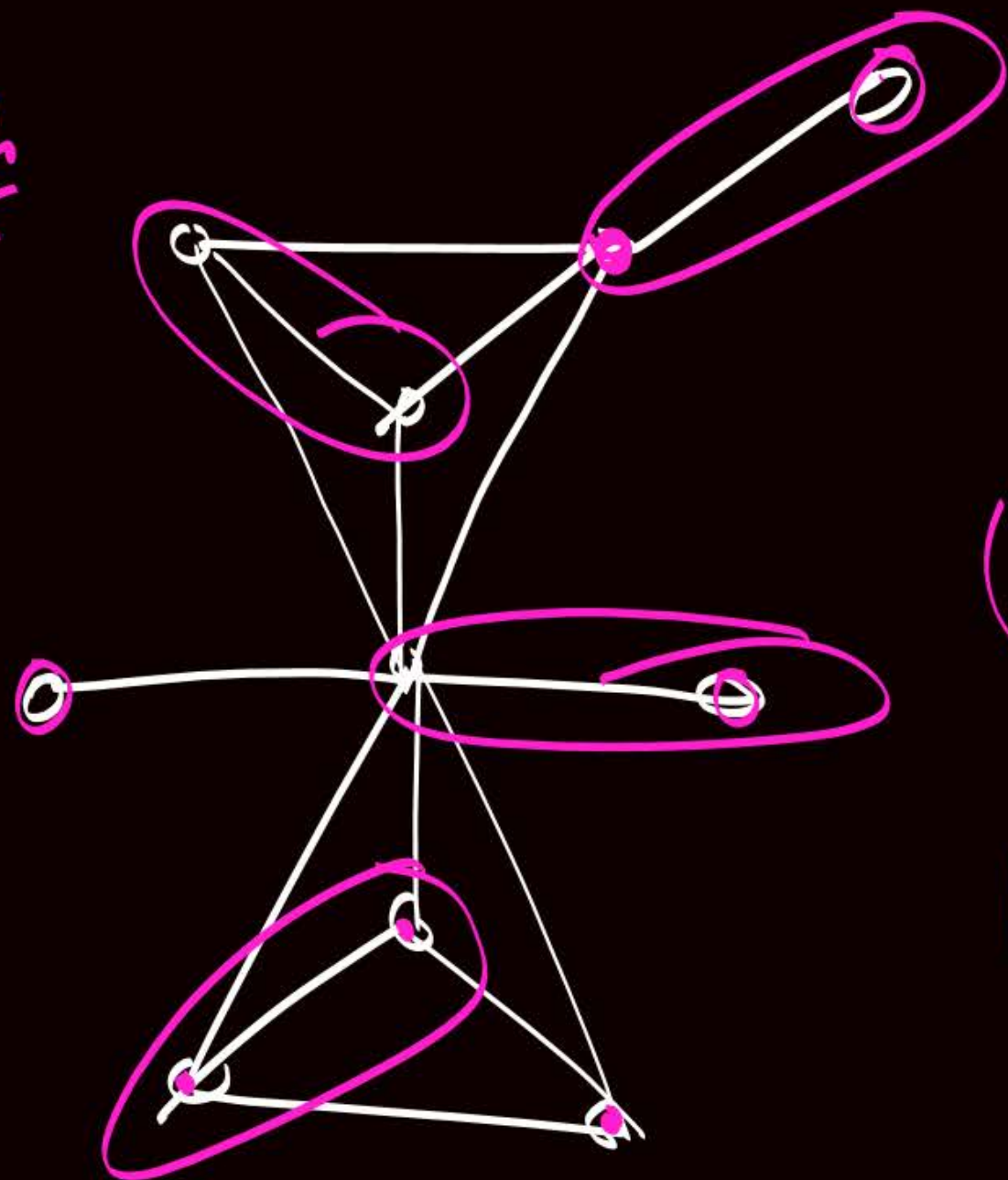


Find  $\alpha_1$  &  $\beta_1$  for given graph

(Line covering No.)  $\alpha_1 = 6$

(Line independence no.)  $\beta_1 =$

Q.10



Find  $\alpha_1$  &  $\beta_1$  for given graph

(Line Covering No.)  $\alpha_1 = 6$

(Line independence no.)  $\beta_1 = (10 - 6) = 4$





## Topic : NOTE



For a graph  $G = (V, E)$

Similar statements  
can not be  
defined w.r.t.  
line covering &  
line independent set.  
and

① If 'S' is a vertex independent set of graph G,  
then 'V-S' is a vertex covering of graph G.

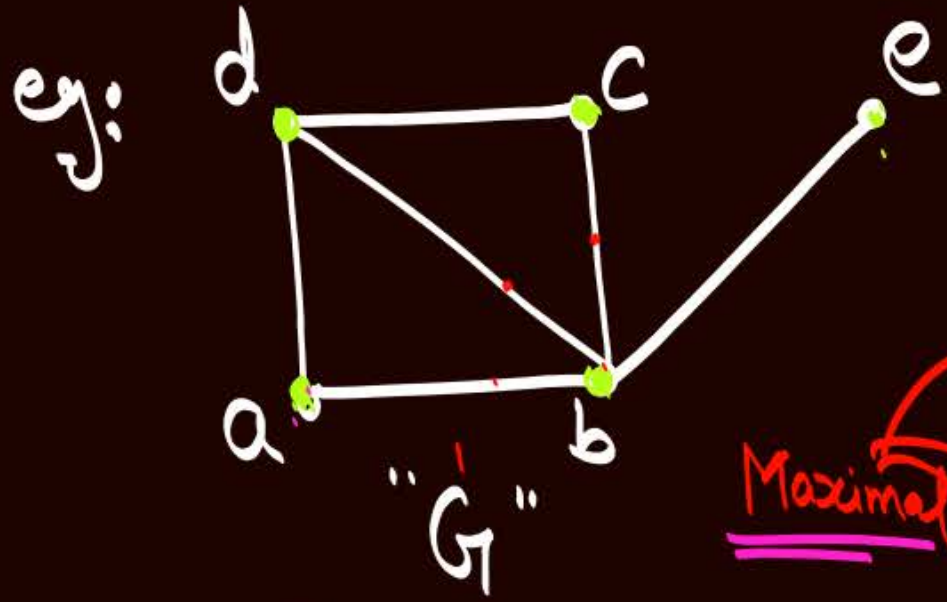
{ if S is maximal then 'V-S' is minimal  
if S is maximum then 'V-S' is minimum }

② If 'K' is a vertex covering of graph 'G', then  
'V-K' is a vertex independent set of graph G.

{ if 'K' is minimal, then 'V-K' is maximal  
if 'K' is minimum, then 'V-K' is maximum }



# Relationship b/w vertex covering & vertex independent set



$G = (V, E)$   
 $V = \{a, b, c, d, e\}$

Vertex independent Set (V-S) = Vertex Covering

$$S_1 = \{e\}$$

$$V - S_1 = \{a, b, c, d\}$$

$$S_2 = \{b\}$$

$$V - S_2 = \{a, c, d, e\}$$

$$S_3 = \{a, e\}$$

$$V - S_3 = \{b, c, d\}$$

$$S_4 = \{a, c, e\}$$

$$V - S_4 = \{b, d\}$$

$$S_5 = \{d, e\}$$

$$V - S_5 = \{a, b, c\}$$

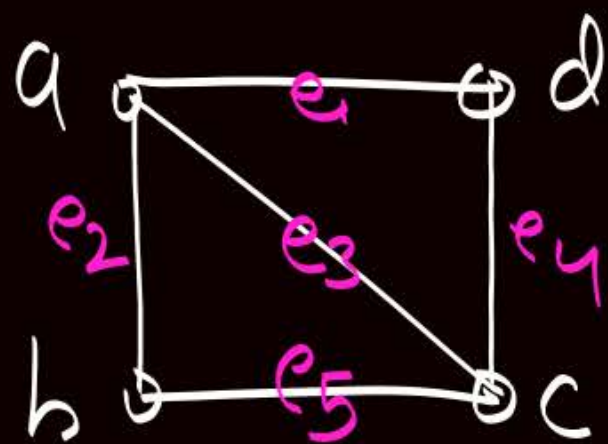
Minimal  
Vertex  
Cover

Minimum

Maximum



Let  $G = (V, E)$  be a graph



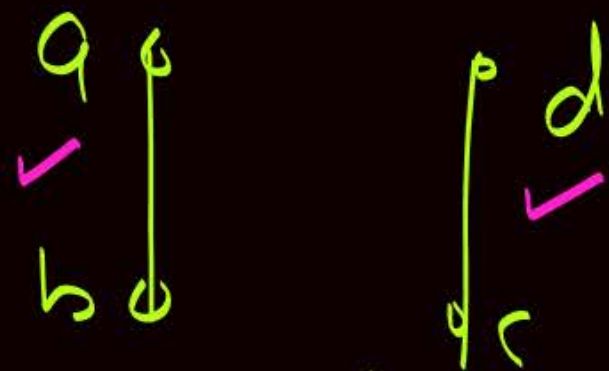
$G = (V, E)$

$V = \{a, b, c, d\}$

$E = \{e_1, e_2, e_3, e_4, e_5\}$

We can observe

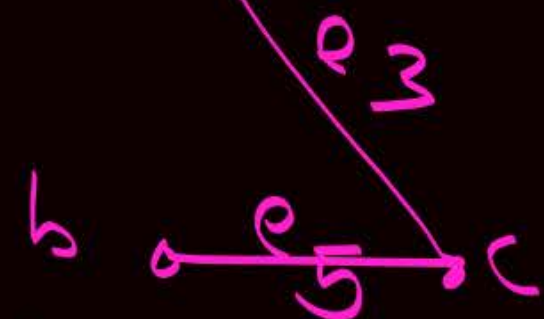
$C_1 = \{e_2, e_4\}$



is a line  
covering

But  $E - C_1 = \{e_1, e_3, e_5\}$

i.e.  $a \xrightarrow{e_1} d$



is not an  
independent set  
of edges.

Connectivity



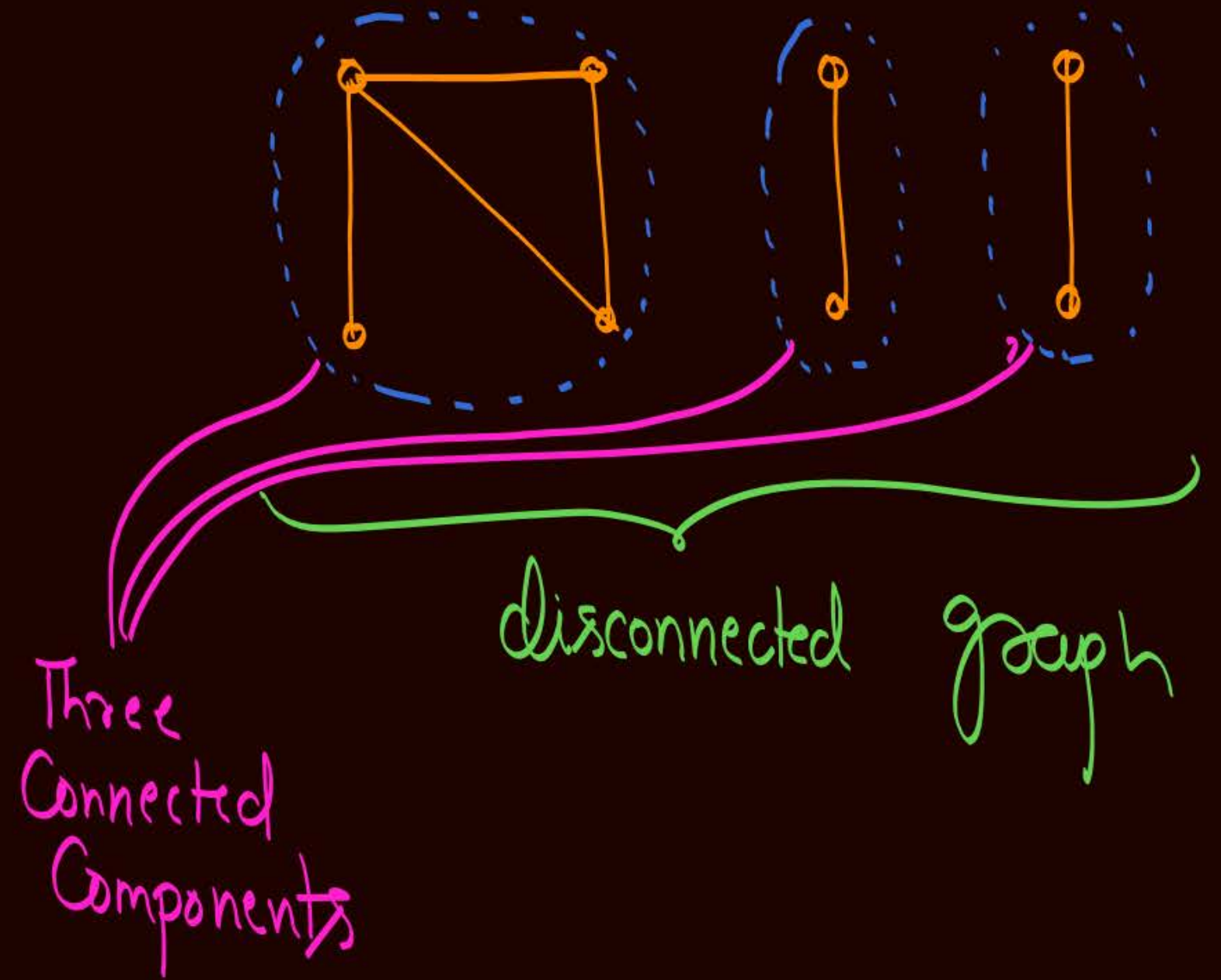
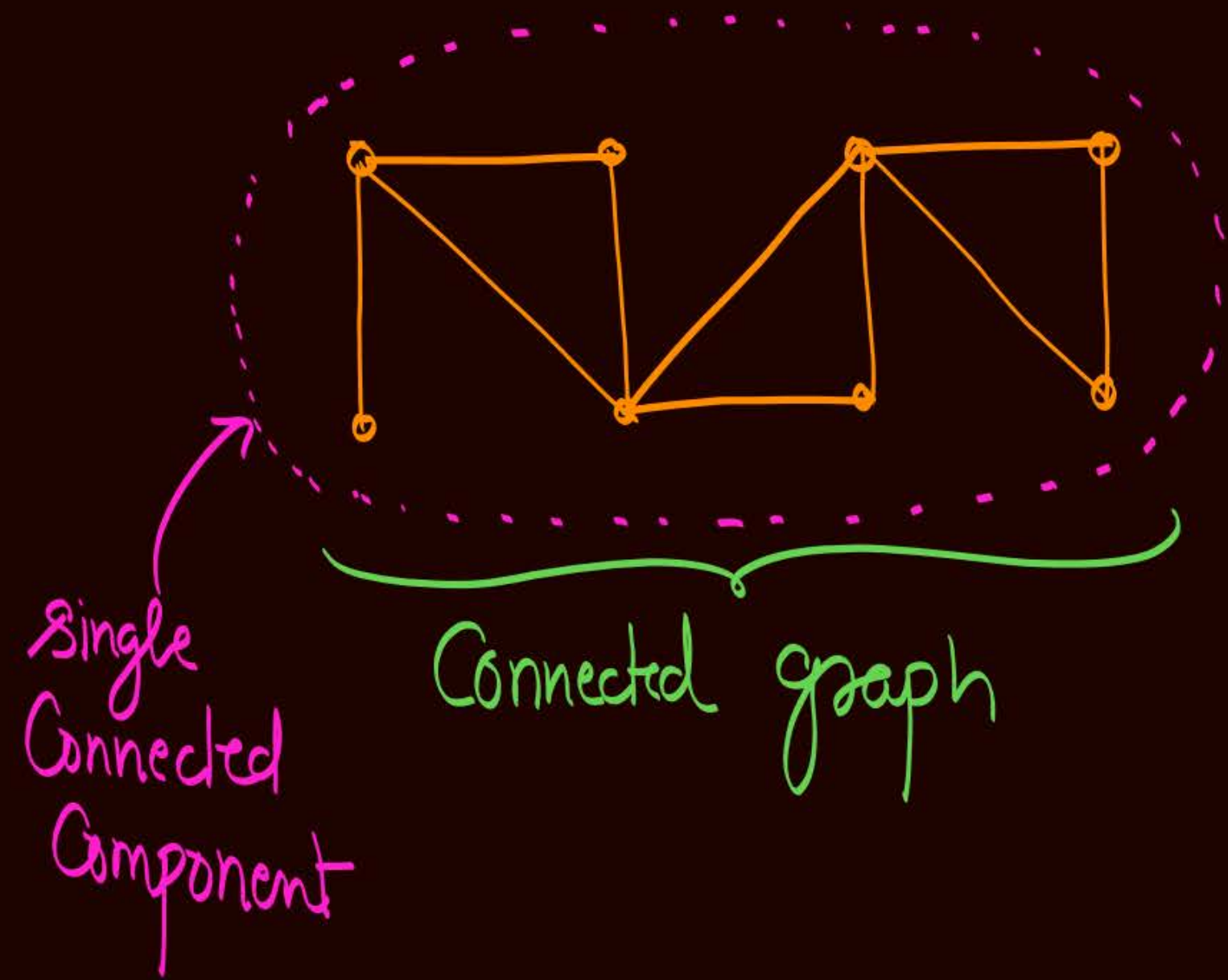


## Topic : Connectivity



A graph is said to be connected if there exists a path between every pair of vertices.

If a graph (with 2 or more vertices) is not connected, then it will have two or more connected components.





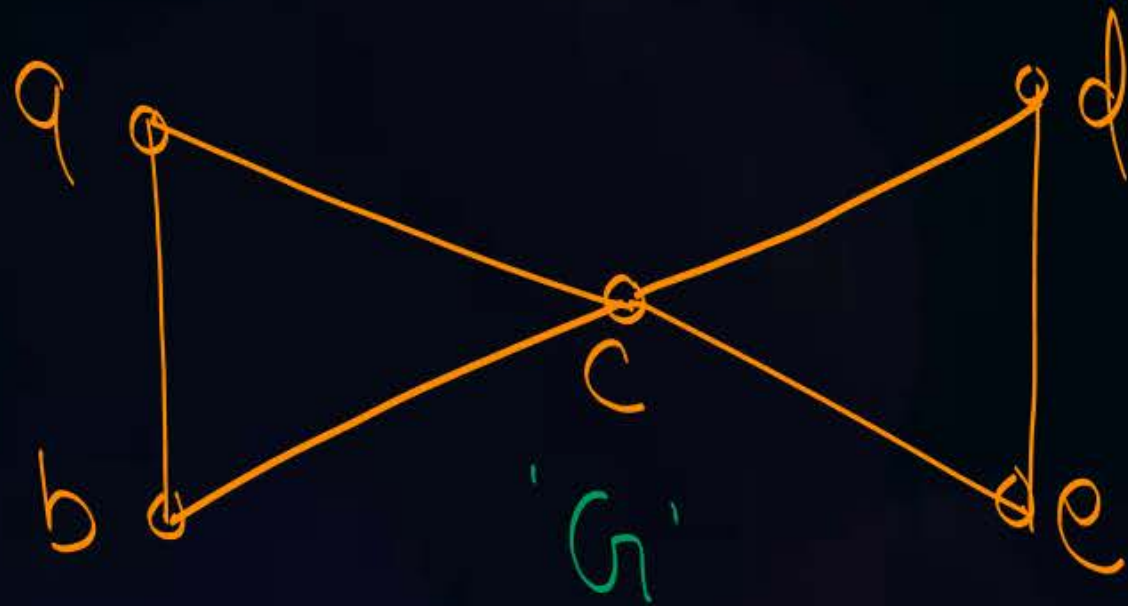


## Topic : Cut-vertex

Articulation point



Let  $G$  be a connected graph, a vertex  $v \in G$  is called a cut vertex if deletion of vertex " $v$ " from graph  $G$  makes the graph disconnected.



it is a connected graph

delete 'c'

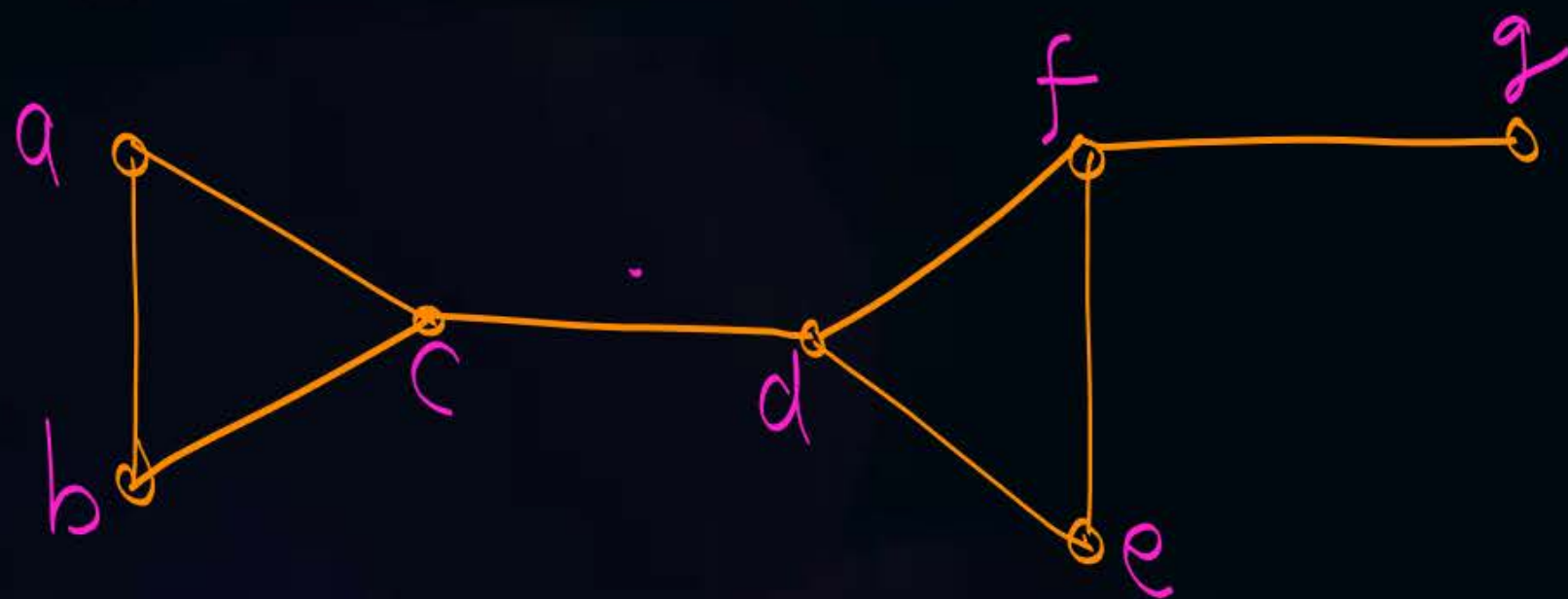


Graph becomes disconnected  
so 'c' is a Cut-vertex



## Topic : Cut-edge / Bridge

Let  $G$  be a connected graph, an edge  $e \in G$  is called a cut edge or bridge if deletion of edge " $e$ " from graph  $G$  makes the graph disconnected.



Connected graph

Edge  $\{c,d\}$  is a Cut edge  
as well as  
Edge  $\{f,g\}$  is also a Cut edge





## Topic : NOTE



① An edge of a connected graph  $G$  can be a cut-edge (bridge) if and only if that edge is not a part of any cycle in the graph.

② At least one vertex associated with a cut-edge is a cut-vertex of that graph

eg. w.r.t. cut-edge  $\{c, d\}$  in above example, both  $c$  &  $d$  are cut-vertices

w.r.t. cut-edge  $\{f, g\}$  in above example, 'g' is not a cut-vertex but 'f' is a cut-vertex

Note:- Existence of Cut-vertex in a graph does not  
imply the existence of cut-edge, but

Existence of a cut-edge in a graph  
implies the existence of cut-vertex.

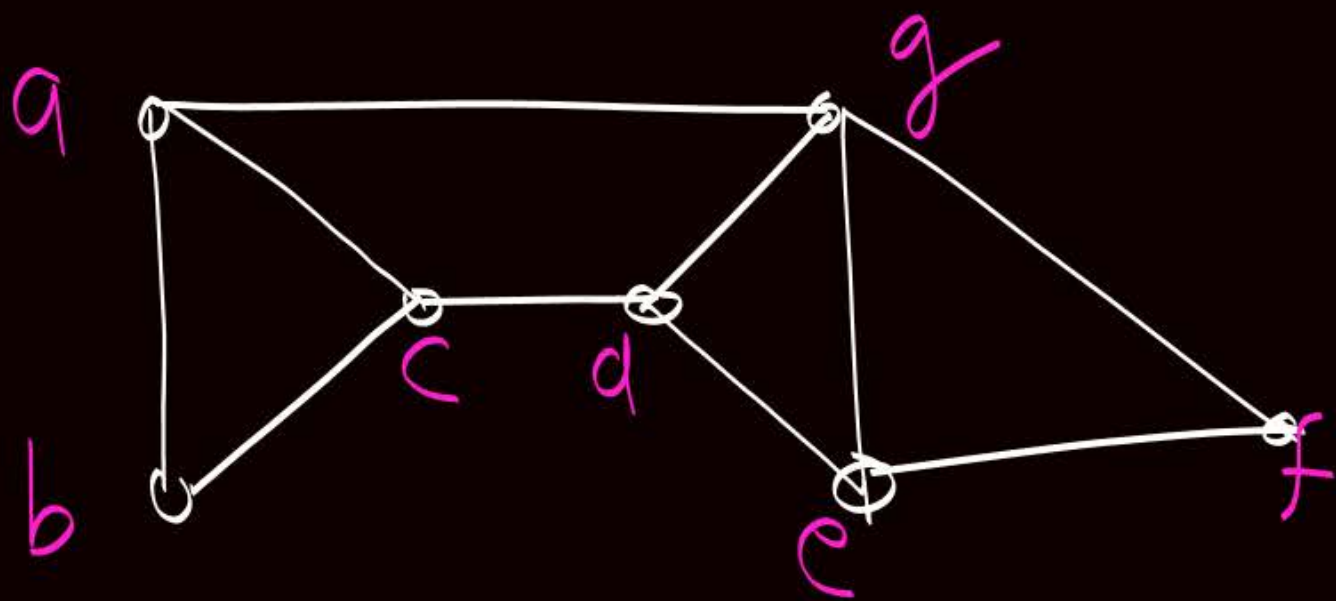




## Topic : Cut-set

A **minimal set** of edges in <sup>a connected</sup> graph  $G$ , such that removal of all the edges of that set from the graph  $G$  makes the graph disconnected is called cut-set of graph  $G$ .

If we leave any edge of that set and delete the remaining edges then it should never disconnect the connected graph.



The set of edges  
 $\{c,d\}, \{d,e\}, \{a,g\}$  is  
 a Cut set or not?

Removal of all the edges  
 will make the graph  
 disconnected, but set is  
 not minimal because  
 edge  $\{d,e\}$  can be removed  
 from the set.  
 so it is not a Cut-set





## Topic : Vertex connectivity

- Let  $G$  be a connected graph, the minimum number of vertices whose removal from graph  $G$  makes the graph disconnected is called vertex connectivity of graph  $G$ . Vertex connectivity of a graph  $G$  is denoted by  $\kappa(G)$
- \* Vertex connectivity of graph is 1 if and only if cut-vertex exists in the graph.
- \* If graph  $G$  is disconnected graph, then  $\kappa(G) = 0$





## Topic : Edge connectivity

Let  $G$  be a connected graph, the minimum number of edges whose removal from graph  $G$  makes the graph disconnected is called edge connectivity of graph  $G$ .  
Edge connectivity of a graph  $G$  is denoted by  $\lambda(G)$

→ Edge connectivity of graph is 1 if and only if cut-edge exists in the graph.

→ If Graph  $G$  is a disconnected graph, then  $\lambda(G) = 0$



Note:-

① For a connected graph  $G$ ,

$$\lambda(G) \leq \delta(G) \text{ --- eq}^n \text{ ①}$$

② For a connected graph  $G$

$$K(G) \leq \lambda(G) \text{ --- eq}^n \text{ ②}$$

By eq<sup>n</sup> ① & ②

$$K(G) \leq \lambda(G) \leq \delta(G)$$

Q:- Find  $\lambda(G)$  and  $K(G)$  for Complete graph  $K_n$ .

•  $\lambda(K_n) =$



Note: If an isolated vertex exists along with other components in the graph, then isolated vertex is treated as a Connected Component

- If there is only '1' isolated vertex then it is Exceptional Case in which it may be treated as Connected or disconnected as per situation

H.W. Find vertex connectivity ( $K(G)$ ) and Edge Connectivity ( $\lambda(G)$ )  
For the following graphs.

|                                      | vertex connectivity | Edge Connectivity |
|--------------------------------------|---------------------|-------------------|
| ① Complete graph $K_n$               |                     |                   |
| ② Cycle graph $C_n$                  |                     |                   |
| ③ Wheel graph $W_n$                  |                     |                   |
| ④ Complete bipartite graph $K_{m,n}$ |                     |                   |
| ⑤ Star graph with $n$ -vertices      |                     |                   |



#Q. In a connected graph G if we delete an edge then no. of components are

**A** 0 or 1

**B** 1 or 2

**C** 2

**D**  $\geq 2$

[MCQ]

H.W.

with 'n' vertices

#Q. In a connected graph G <sup>↑</sup>if we delete a vertex then number of components are

**A**  $\leq 2$

**B** 1 or 2

**C** Lies between 1 and  $n - 1$

**D**  $\geq 2$



H.W.



#Q. A simple graph with  $n$  vertices is necessarily connected if number of edges are more than  $E$ , then find the value of  $E$ .

H.W.



#Q. Maximum number of edges possible in <sup>a simple</sup>  $\wedge$  undirected graph with  $n$ -vertices and  $k$  components?





## 2 mins Summary



✓ **Topic**

Vertex covering

✓ **Topic**

Vertex independent set

✓ **Topic**

Connectivity

**THANK - YOU**